

Skema Database (Database Schema) - RubbyBooks

Database dirancang dengan skema relasional yang terdiri dari lebih dari 8 tabel wajib (total ada 14 tabel pendukung).

1. Tabel Utama Pengguna & Kategori

users

Tabel yang menampung semua entitas pengguna aplikasi.

- `id` (INT, PK, Auto Increment)
- `name` (VARCHAR)
- `email` (VARCHAR, Unique)
- `password` (VARCHAR, Hashed)
- `role` (ENUM: 'buyer', 'seller', 'admin')
- `status` (ENUM: 'active', 'pending', 'banned', 'suspended')
- `avatar` (VARCHAR)
- `phone` (VARCHAR)
- `dob` (DATE)
- `alternate_phone` (VARCHAR)
- `delete_requested` (TINYINT)
- `created_at` (TIMESTAMP)

seller_verifications

Tabel khusus untuk mengelola proses verifikasi pengguna dengan *role* seller.

- `id` (INT, PK)
- `seller_id` (INT, FK -> users.id)
- `status` (ENUM: 'pending', 'approved', 'rejected')
- `approved_by` (INT, FK -> users.id)
- `approved_at` (TIMESTAMP)

categories

Tabel untuk mengelompokkan buku.

- `id` (INT, PK)
- `name` (VARCHAR)

- description (TEXT)

2. Tabel Produk & Aktivitas Belanja

products

Menampung seluruh katalog buku.

- id (INT, PK)
- seller_id (INT, FK -> users.id)
- category_id (INT, FK -> categories.id)
- name (VARCHAR)
- description (TEXT)
- price (INT)
- stock (INT)
- book_condition (ENUM: 'new', 'used_good', 'used_fair')
- image (VARCHAR)
- status (ENUM: 'active', 'inactive')
- created_at (TIMESTAMP)

wishlists

Daftar buku favorit pembeli.

- id (INT, PK)
- buyer_id (INT, FK -> users.id)
- product_id (INT, FK -> products.id)
- created_at (TIMESTAMP)

carts

Keranjang belanja pembeli.

- id (INT, PK)
- buyer_id (INT, FK -> users.id)
- product_id (INT, FK -> products.id)
- qty (INT)

3. Tabel Pesanan & Transaksi

shipping_addresses

Alamat pengiriman terkait setiap pesanan.

- `id` (INT, PK)
- `buyer_id` (INT, FK -> `users.id`)
- `recipient_name` (VARCHAR)
- `phone` (VARCHAR)
- `address` (TEXT)
- `city` (VARCHAR)
- `postal_code` (VARCHAR)

orders

Tabel untuk setiap *checkout* transaksi.

- `id` (INT, PK)
- `buyer_id` (INT, FK -> `users.id`)
- `shipping_address_id` (INT, FK -> `shipping_addresses.id`)
- `invoice_number` (VARCHAR, Unique)
- `total` (INT)
- `shipping_cost` (INT)
- `receipt_number` (VARCHAR)
- `status` (ENUM: 'pending', 'paid', 'processing', 'shipped', 'delivered', 'cancelled')
- `created_at` (TIMESTAMP)

order_items

Rincian item/produk dalam sebuah order.

- `id` (INT, PK)
- `order_id` (INT, FK -> `orders.id`)
- `product_id` (INT, FK -> `products.id`)
- `qty` (INT)
- `price` (INT)
- `subtotal` (INT)

payments

Tabel pembayaran/upload bukti transfer.

- `id` (INT, PK)
- `order_id` (INT, FK -> `orders.id`)
- `method` (VARCHAR)
- `proof` (VARCHAR)
- `status` (ENUM: 'waiting', 'accepted', 'rejected')
- `paid_at` (TIMESTAMP)

4. Tabel Fitur Pendukung

reviews

Sistem ulasan dan pemberian bintang untuk produk.

- `id` (INT, PK)
- `product_id` (INT, FK -> `products.id`)
- `buyer_id` (INT, FK -> `users.id`)
- `rating` (TINYINT)
- `comment` (TEXT)
- `seller_reply` (TEXT)
- `photo` (VARCHAR)
- `created_at` (TIMESTAMP)

notifications

Sistem notifikasi dalam aplikasi (*in-app notification*).

- `id` (INT, PK)
- `user_id` (INT, FK -> `users.id`)
- `message` (TEXT)
- `is_read` (TINYINT)
- `created_at` (TIMESTAMP)

system_logs

Catatan rekaman aktivitas/log dari pengguna atau sistem.

- `id` (INT, PK)

- activity (TEXT)
- created_at (TIMESTAMP)

system_settings

Menyimpan konfigurasi pengaturan aplikasi.

- key (VARCHAR, PK)
- value (TEXT)

5. Diagram ERD (PlantUML)

[!NOTE] Jika ekstensi PlantUML Anda memunculkan error *missing @start/@end wrapper* saat melihat berkas Markdown ini, silakan buka dan render langsung berkas standalone [DATABASE_SCHEMA.puml](#).

Berikut adalah kode PlantUML untuk merepresentasikan hubungan antar tabel (Entity Relationship Diagram):

```
@startuml
!theme plain
hide circle
skinparam linetype ortho

entity "users" {
    * id : INT <<PK>>
    --
    * name : VARCHAR
    * email : VARCHAR <<Unique>>
    * password : VARCHAR
    * role : ENUM
    * status : ENUM
    avatar : VARCHAR
    phone : VARCHAR
    dob : DATE
    alternate_phone : VARCHAR
    delete_requested : TINYINT
    created_at : TIMESTAMP
}

entity "seller_verifications" {
    * id : INT <<PK>>
    --
    * seller_id : INT <<FK>>
    * status : ENUM
    approved_by : INT <<FK>>
    approved_at : TIMESTAMP
}

entity "categories" {
    * id : INT <<PK>>
    --
    * name : VARCHAR
    description : TEXT
}

entity "products" {
    * id : INT <<PK>>
```

```

--
* seller_id : INT <<FK>>
* category_id : INT <<FK>>
* name : VARCHAR
description : TEXT
* price : INT
* stock : INT
* book_condition : ENUM
image : VARCHAR
* status : ENUM
created_at : TIMESTAMP
}

entity "wishlists" {
  * id : INT <<PK>>
  --
  * buyer_id : INT <<FK>>
  * product_id : INT <<FK>>
  created_at : TIMESTAMP
}

entity "carts" {
  * id : INT <<PK>>
  --
  * buyer_id : INT <<FK>>
  * product_id : INT <<FK>>
  * qty : INT
}

entity "shipping_addresses" {
  * id : INT <<PK>>
  --
  * buyer_id : INT <<FK>>
  * recipient_name : VARCHAR
  * phone : VARCHAR
  * address : TEXT
  * city : VARCHAR
  * postal_code : VARCHAR
}

entity "orders" {
  * id : INT <<PK>>
  --
  * buyer_id : INT <<FK>>
  shipping_address_id : INT <<FK>>
  * invoice_number : VARCHAR <<Unique>>
  * total : INT
  * shipping_cost : INT
  receipt_number : VARCHAR
  * status : ENUM
  created_at : TIMESTAMP
}

entity "order_items" {
  * id : INT <<PK>>
  --
  * order_id : INT <<FK>>
  * product_id : INT <<FK>>
  * qty : INT
  * price : INT
  * subtotal : INT
}

entity "payments" {

```

```

* id : INT <<PK>>
--
* order_id : INT <<FK>>
* method : VARCHAR
proof : VARCHAR
* status : ENUM
paid_at : TIMESTAMP
}

entity "reviews" {
* id : INT <<PK>>
--
* product_id : INT <<FK>>
* buyer_id : INT <<FK>>
* rating : TINYINT
comment : TEXT
seller_reply : TEXT
photo : VARCHAR
created_at : TIMESTAMP
}

entity "notifications" {
* id : INT <<PK>>
--
* user_id : INT <<FK>>
* message : TEXT
* is_read : TINYINT
created_at : TIMESTAMP
}

entity "system_logs" {
* id : INT <<PK>>
--
* activity : TEXT
created_at : TIMESTAMP
}

entity "system_settings" {
* key : VARCHAR <<PK>>
--
* value : TEXT
}

users ||--o{ seller_verifications : "seller_id (FK)"
users ||--o{ seller_verifications : "approved_by (FK)"
users ||--o{ products : "seller_id (FK)"
categories ||--o{ products : "category_id (FK)"
users ||--o{ wishlists : "buyer_id (FK)"
products ||--o{ wishlists : "product_id (FK)"
users ||--o{ carts : "buyer_id (FK)"
products ||--o{ carts : "product_id (FK)"
users ||--o{ shipping_addresses : "buyer_id (FK)"
users ||--o{ orders : "buyer_id (FK)"
shipping_addresses ||--o{ orders : "shipping_address_id (FK)"
orders ||--o{ order_items : "order_id (FK)"
products ||--o{ order_items : "product_id (FK)"
orders ||--|| payments : "order_id (FK)"
products ||--o{ reviews : "product_id (FK)"
users ||--o{ reviews : "buyer_id (FK)"
users ||--o{ notifications : "user_id (FK)"

```

@endum

